

REPRO Report

[Recorded AI trace + selected checkpoints] Color hex readback changes

Case md-12623-terra-replay-02 | Mindustry | Awaiting human
Revision: ef5d3cb26bcd794a6e560dbb8d20b6f8200b130c
Reported platform: Windows (reported); Linux (planned evaluation) | Build: Unspecified
Generated 2026-09-14 21:43 UTC from the current saved case.

Candidate ready for review. Validation is incomplete or failed; inspect the recorded checks before using the patch.

Reproduction	Time to first proof	Replay actions
5/5 clean runs	9:16	29 to 29

Player report

Platforms

Windows

Build

v159.7

Issue

'Colored Wall' & 'Colored Floor' tiles in editor do not accept every hex color code given to them in game. I tested this on both v159.7 and BE 27795 just to make sure it was not yet addressed.

In the save file below I did it for the following values: ff0000 to ff2000. Here is the table for the hex codes inputted vs accepted by the editor so you can copy paste:

ff0000 -> ff0000
ff0100 -> ff0100
ff0200 -> ff0200
ff0300 -> ff0200
ff0400 -> ff0300
ff0500 -> ff0500
ff0600 -> ff0600
ff0700 -> ff0700
ff0800 -> ff0700
ff0900 -> ff0800
ff0a00 -> ff0a00
ff0b00 -> ff0b00
ff0c00 -> ff0b00
ff0d00 -> ff0c00
ff0e00 -> ff0e00
ff0f00 -> ff0f00
ff1000 -> ff1000
ff1100 -> ff1000
ff1200 -> ff1100
ff1300 -> ff1300
ff1400 -> ff1400
ff1500 -> ff1400
ff1600 -> ff1500
ff1700 -> ff1600
ff1800 -> ff1800
ff1900 -> ff1900
ff1a00 -> ff1900
ff1b00 -> ff1a00
ff1c00 -> ff1c00
ff1d00 -> ff1d00
ff1e00 -> ff1e00
ff1f00 -> ff1e00
ff2000 -> ff1f00

Steps to reproduce

1. Open the map editor and go to edit any map
2. Select the 'Colored Floor' or 'Colored Wall' tile in editor

3. Paste in any hex color codes that differ by 1 in value and place each one down next to one another, eventually you'll see that the tiles no longer separate as the hex color code did not change its value properly

```

```

Mods used

none

Save file

[Debris Field With Color Tiles For You To Look
At.zip](https://github.com/user-attachments/files/32078489/Debris.Field.With.Color.Tiles.For.You.To.Look.At.zip)

(Crash) logs

No response

Explicit REPRO assistance for this separate qualification job

The prior report-only Terra investigator executed a clean final 29-action sequence after a reset, but its final model response exceeded its output allowance before returning a complete proposal. This variant supplies those 29 unchanged actions (original event action indices 55-83), cached report triage, and six existing entered/confirmed/readback checkpoint labels selected by an operator. No historical developer fix or source diagnosis is supplied. The fresh verification scope is ff0300 readback for a placed-and-picked Colored Wall and for the Colored Floor picker after confirmation/reopening; it does not cover the entire reported value matrix or adjacent floor placement. Neither the old investigator hypothesis nor old screenshots count as fresh confirmations. Five new clean-profile confirmations are required before source diagnosis or patching. The new worker paces wheel inputs, but this selected final trace contains no scrolling.

Expected behavior: Per the reporter's expectation, each supported six-character hexadecimal color value entered for a Colored Floor or Colored Wall tile should be retained/read back as that same value, so separately entered values remain distinct.

Initial questions

Recorded at intake; later findings and validation may address these questions.

- The report has not been independently reproduced or verified; it does not by itself establish that a defect exists.
- It is unspecified how the reported "accepted" value was determined: editor-field readback, visible tile appearance, serialized save data, or another method.
- No result matrix distinguishes Colored Floor from Colored Wall, so it is unclear whether every listed mapping was observed for both tile types.
- No result matrix distinguishes v159.7 from "BE 27795", so it is unclear whether the same mappings occurred in each stated build.
- The exact entry workflow is not provided, including the specific control used, whether the listed table was produced by paste or typing, whether a # prefix was used, required confirmation/focus changes, and when the value was inspected.
- The map type, map settings, tile coordinates, and correspondence between placed tiles and table entries are not specified.

- Build identity is unresolved: the report body names v159.7 and "BE 27795"; metadata names commit ef5d3cb26bcd794a6e560dbb8d20b6f8200b130c; build_version is null. Their relationship is not supplied.
- Linux is described only as planned evaluation in metadata; no Linux observation is reported.
- Documentation or UI rules establishing that the listed syntax and exact value fidelity are supported are not provided.

Recorded reproduction

1. Open the map editor from the clean-profile main menu.
2. Open the new-map dialog for an isolated color-input test.
3. Focus the test-map name field.
4. Name the isolated test map.
5. Create the isolated editable map.
6. Focus the editor tile search field.
7. Filter tile selection to Colored Wall.
8. Select the filtered Colored Wall tile.
9. Open the Colored Wall color picker.
10. Focus the hexadecimal color field.
11. Select the existing color value.
12. Enter the reported six-character hexadecimal color ff0300.
13. Confirm the entered ff0300 color.
14. Select the editor pencil tool for placement.
15. Place a Colored Wall using the confirmed ff0300 input.
16. Switch to the editor pick tool.
17. Pick the placed Colored Wall so its stored configuration becomes the editor selection.
18. Open the placed wall's picked color value for readback.
19. Close the wall readback dialog.
20. Focus the tile search field to switch tile type.
21. Select the existing tile search text.
22. Filter the editor to Colored Floor.
23. Select the filtered Colored Floor tile.
24. Open the selected Colored Floor color picker.
25. Focus the floor hexadecimal color field.
26. Select the existing floor color value.
27. Enter the same reported six-character hexadecimal color for Colored Floor.
28. Confirm the Colored Floor color input.
29. Reopen the floor picker to read back its committed value.

Source diagnosis

Most likely localization is the hexadecimal-input path in ColorPicker: after parsing the entered text into `current`, it immediately derives HSV and reconstructs `current` from HSV before OK submits it. That RGBHSVRGB round trip occurs before the Colored Floor/Wall caller packs the color to RGBA8888. It is therefore the shared source-level path capable of changing `ff0300` before either tile type stores or rereads it; the precise one-channel rounding behavior inside Arc's unavailable Color implementation is not directly inspectable here.

core/src/minindustry/ui/dialogs/ColorPicker.java

The recorded replay reports the same ff0300-to-ff0200 readback for both Colored Wall and Colored Floor. This is the only inspected shared code path that parses typed hexadecimal input and mutates the Color object before either tile-specific callback receives it.

core/src/minindustry/world/blocks/environment/ColoredFloor.java

This is the common integration point for Colored Floor and Colored Wall and packs the post-picker floating Color to the persisted integer. It is a secondary candidate because packing can expose a sub-8-bit difference introduced by the picker, while the inspected tile storage/pick methods themselves appear to preserve the integer.

Proposed patch

Why this patch should work

Remove the HSV-to-RGB reconstruction immediately after `Color.valueOf(current, value)` parses hex text. The parsed `current` color is already exact; the callback still updates the HSV state and slider positions for UI consistency. Consequently, pressing OK without changing a slider sends the parsed RGB value to the Colored Floor/Wall callback, which then packs and stores that value unchanged.

Risks and tradeoffs

- This changes hex-entry behavior for every caller of the shared ColorPicker, not only colored environment tiles. That scope is intentional because the lossy conversion is in the shared picker path.
- After direct hex entry, subsequent slider interaction still intentionally derives a color from HSV and may quantize according to slider precision; this patch only preserves exact text input when no slider edit is made.
- Validation should exercise six- and eight-digit input, alpha-enabled and alpha-disabled picker callers, invalid/intermediate text, direct OK, reopen/readback, and slider edits after text entry.

Code changes

Lines starting with - are removed; lines starting with + are added. Unmarked lines provide context.

```
diff --git a/core/src/minindustry/ui/dialogs/ColorPicker.java b/core/src/minindustry/ui/dialogs/ColorPicker.java
index f5e096c..35ff0c8 100644
--- a/core/src/minindustry/ui/dialogs/ColorPicker.java
+++ b/core/src/minindustry/ui/dialogs/ColorPicker.java
@@ -146,7 +146,6 @@ public class ColorPicker extends BaseDialog{
     aSlider.setValue(a);
 }

-    updateColor(false);
-} catch (Exception ignored){
-}
-}).size(130f, 40f).valid(text -> {
```

Validation results

Validation is incomplete or has failed. The recorded results below determine what has been verified.

Check	Result	Recorded evidence
Regression before patch	PASS	Bug oracle triggered in 5/5 clean runs; the regression therefore fails on the pre-fix build.
Candidate build	PASS	Build exit code 0
Existing tests	PASS	Test exit code 0.
Original replay after patch	FAIL	0/5 reached expected state without the symptom; bug seen 5 times.
Smoke test	PASS	Clean desktop launch and live process after startup; deeper gameplay smoke coverage is not implemented.

Source analysis limitations

These are the source reviewer's recorded notes. That stage inspects code; see Recorded reproduction and Validation results for executed game checks.

- This was a source-only review. I did not execute the game or inspect the listed image artifacts in this stage. The supplied replay is recorded evidence of 5 successful out of 5 deterministic runs at the supplied commit, not a run performed here.
- The Arc dependency implementation of `Color.valueOf`, `Color.toHsv`, `Color.fromHsv`, and `Color.rgba8888` was not present in the inspectable repository search results. Consequently, source review establishes the lossy-looking round-trip location but cannot independently identify the exact floating-point, slider, or integer-packing operation responsible for the one-value decrement.
- The supplied replay specifically establishes the ff0300 workflow for both tile types; this review did not independently validate every additional reported table entry, Windows/build comparisons, save/archive behavior, or screenshot contents.
- No conclusion is offered about a future fix; the inspected code only localizes the currently recorded behavior.

Usage and review

Each replay starts with a fresh profile. Reproduction counts describe this case; action reduction is bounded and is not proof of a globally minimal replay.

44 model calls; 520,345 input tokens; 95,683 output tokens. Usage accumulates across jobs on this case. PDF export uses saved data and makes no model calls.

The proposed patch lives in a disposable game workspace. Handoff approval records a local review decision and does not publish or merge changes upstream. Visual checks are model-based; startup smoke does not establish broad gameplay coverage.

Investigation timeline

All timestamps are UTC. First and latest entries are saved event times, not active work durations. Repeated attempts and pauses are retained. Selected milestones appear below; the workspace has the full searchable log.

Triage / 2 records

First entry: 14 Sep 2026 20:58:23 UTC

Latest entry: 14 Sep 2026 20:58:23 UTC

14 Sep 2026 20:58:23 | Recorded AI trigger supplied with explicit operator checkpoint selection; not yet freshly verified.

14 Sep 2026 20:58:23 | Operator selected the final 29 recorded actions after reset (original indices 55-83) and six existing color-input/readback labels. No action parameters changed; cached triage is explicit. No retained images count as fresh...

Reproduce / 193 records

First entry: 14 Sep 2026 20:58:23 UTC

Latest entry: 14 Sep 2026 21:22:21 UTC

14 Sep 2026 20:58:23 | Running fresh baseline assertions on the new worker; compiled historical JAR and dependency cache were reused with explicit provenance.

14 Sep 2026 21:22:21 | Symptom observed. The required Wall and Floor picker paths both reach readable reopened-picker states. Each visibly receives `ff0300`, is dismissed through the confirmation transition, and later reads back `ff0200ff` rather than...

Reduce / 1 records

First entry: 14 Sep 2026 21:22:21 UTC

Latest entry: 14 Sep 2026 21:22:21 UTC

14 Sep 2026 21:22:21 | Unchanged supplied color trace confirmed 5/5 with all eight checkpoint images. Explicit assistance and the earlier inconclusive selection remain recorded.

Localize / 48 records

First entry: 14 Sep 2026 21:22:21 UTC

Latest entry: 14 Sep 2026 21:24:51 UTC

14 Sep 2026 21:22:21 | Tracing observed behavior into the pre-fix source.

14 Sep 2026 21:24:51 | Saving the confirmed replay as an executable regression test.

Validate / 184 records

First entry: 14 Sep 2026 21:24:51 UTC

Latest entry: 14 Sep 2026 21:41:36 UTC

14 Sep 2026 21:24:51 | Designing a small candidate diff in the disposable repository.

14 Sep 2026 21:41:36 | Fix not established. The required wall and floor readback paths are visibly reached. Each picker shows the literal six-digit input `ff0300` for the correctly identified tile type, then visibly closes to that type's editor panel....

Review / 1 records

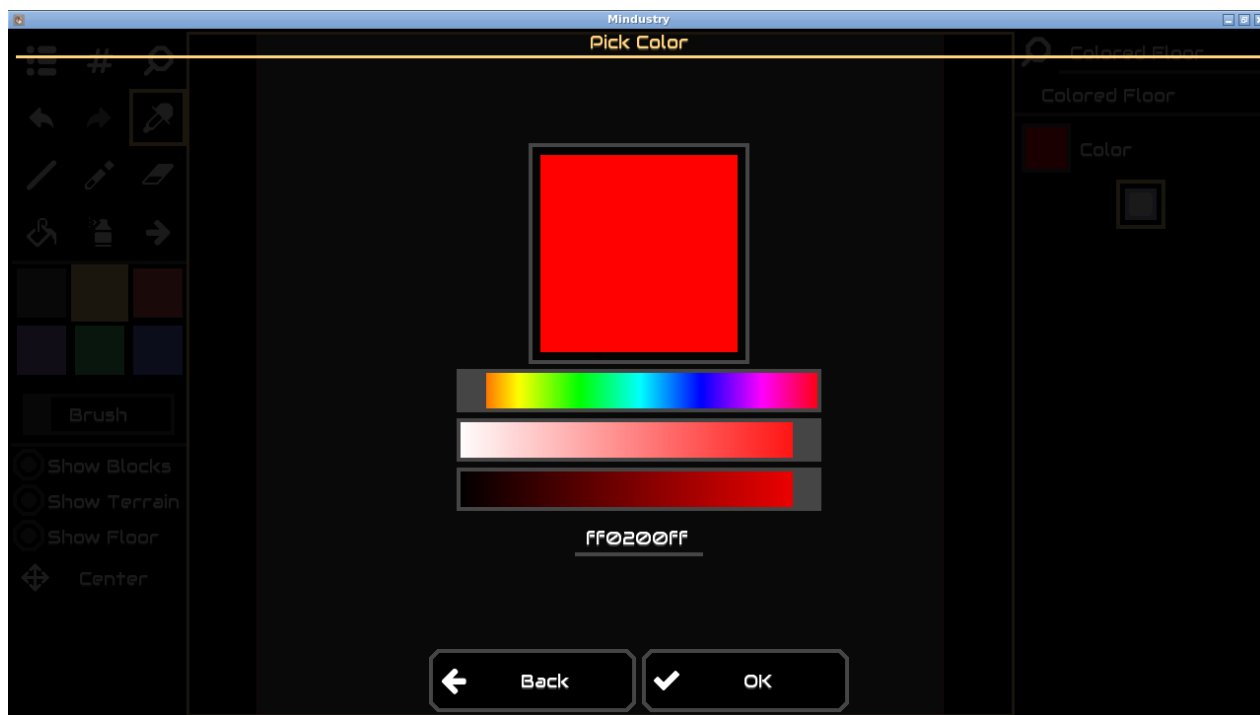
First entry: 14 Sep 2026 21:41:47 UTC

Latest entry: 14 Sep 2026 21:41:47 UTC

14 Sep 2026 21:41:47 | Candidate ready for review. Validation is incomplete or failed; inspect the recorded checks before using the patch.

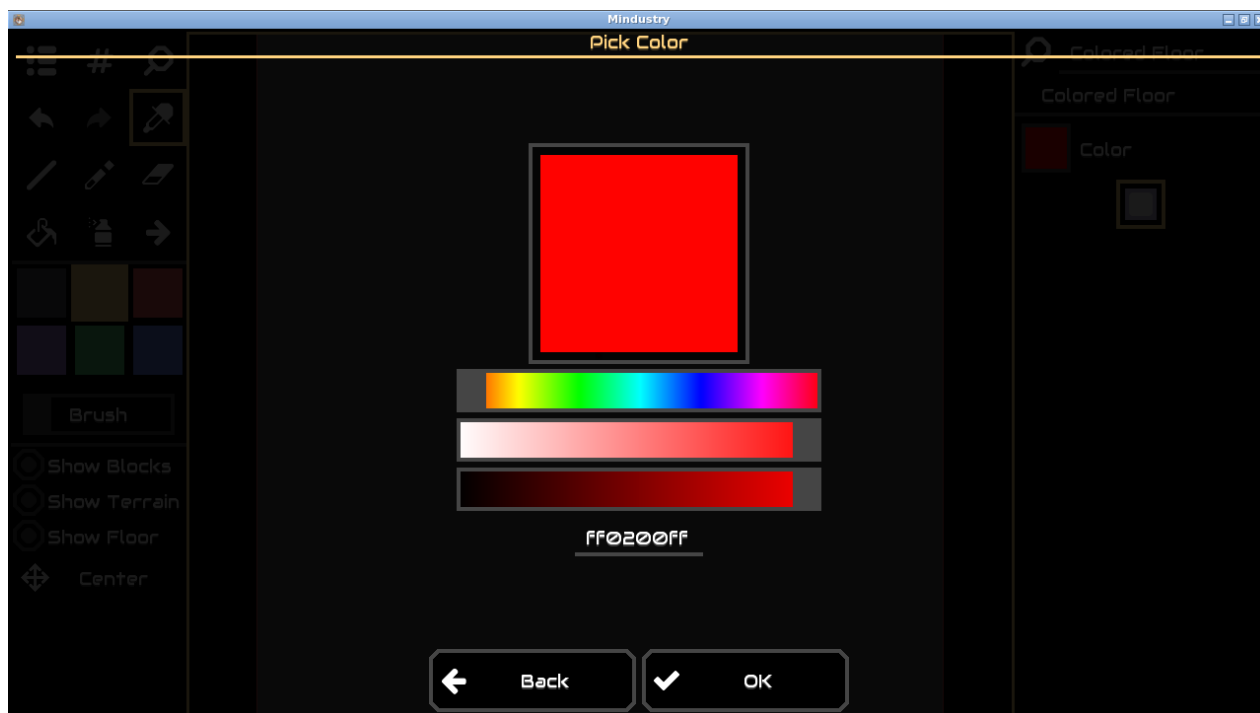
Recorded screenshots

Baseline replay evidence



md-12623-terra-replay-02-b0469bd001d943f79de5-complete-color-confirmation.png

Candidate replay outcome



md-12623-terra-replay-02-b0469bd001d943f79de5-post-patch.png